Cell Biology & Biochemistry

Topics Covered

Biochemistry

Biomolecules and Molecular Organization
Biological Macromolecules:
- Carbohydrates
- Lipids
- Proteins
- Nucleic acids and Nucleotides
- Identification of Biomolecules (Practical)

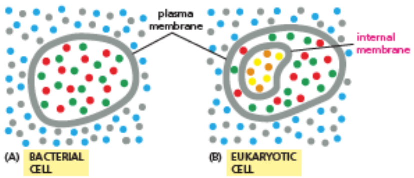
Cell Biology

Cell ultra-structure
Biological membranes: Structure and functions
Cell Division
DNA Replication
Protein Biosynthesis
Cellular communication
Bioenergetics
- Enzymes: General properties and mechanism of action
- Cellular respiration

Cell Biology

Biological Membranes

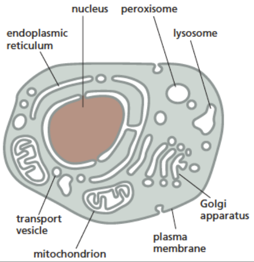
- Membranes define the boundary between cells and the exterior
- Serve as a selective Barrier
- All cells possess cell membranes



Cell Biology

Biological Membranes

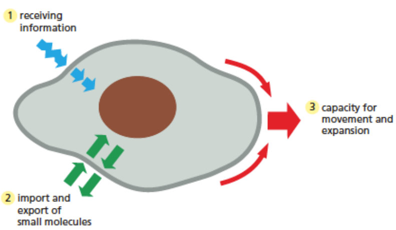
- Cell's internal membranes are also biological membranes
- Organelle membranes also have the same structure



Cell Biology

Biological Membranes

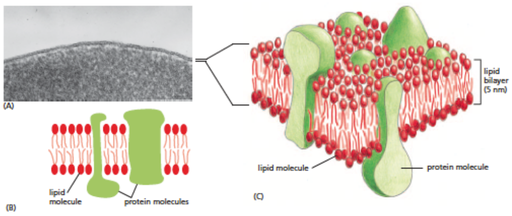
- Cell's transactions with the environment takes place through the cell membrane



Cell Biology

Biological Membranes

- Biological membranes are dynamic structures composed of a diverse set of phospholipid molecules and proteins



Cell Biology

Biological Membranes

Fluid Mosaic model of the plasma membrane

- Proposed by Seymour Singer & Garth Nicholson in 1972 based on experimental evidence
- Mosaic of proteins floating in the fluid lipid bayer
- Components of the cell membrane
  - Phospholipid bilayer
  - Membrane proteins
  - Network of supporting fibers
  - Exterior proteins and glycolipids

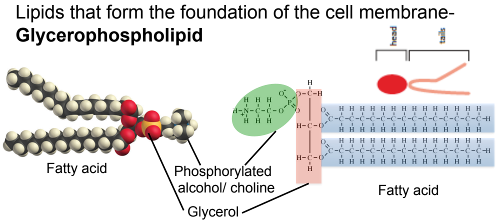
Cell Biology

Biological Membranes

Fluid Mosaic model of the plasma membrane

Phospholipid bilayer

- Membranes of all living cells are made of sheets of lipid only two molecules thick
- Lipids that form the foundation of the cell membrane- **Glycerophospholipid**



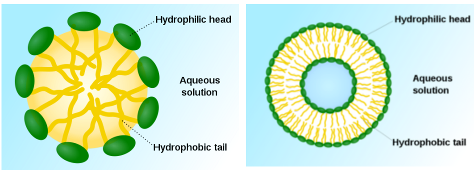
Cell Biology

Biological Membranes

Fluid Mosaic model of the plasma membrane

Phospholipid bilayer

- Phosphorylated alcohol/choline group - strongly polar/water soluble (Hydrophilic)
- Glycerol & Fatty acid region - strongly nonpolar/water-insoluble (Hydrophobic)



Cell Biology

Biological Membranes

Fluid Mosaic model of the plasma membrane

Phospholipid bilayer

- Non-polar interior of the phospholipid bilayer prevents the entry of any water-soluble substances to the cell
- Phospholipid bilayer is fluid – individual phospholipid molecules and proteins are able to move freely

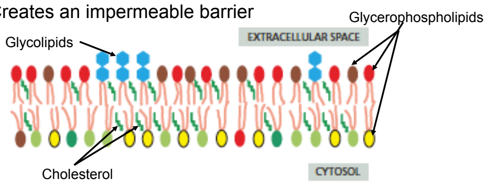
Cell Biology

Biological Membranes

Fluid Mosaic model of the plasma membrane

Phospholipid bilayer

- Every biological membrane is composed of a glycerophospholipid bilayer
- Other components (membrane proteins, cholesterol) are embedded in the bilayer
- Provides a flexible matrix
- Creates an impermeable barrier



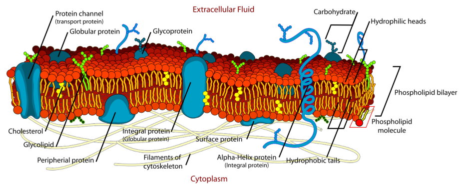
Cell Biology

Biological Membranes

Fluid Mosaic model of the plasma membrane

Network of supporting fibers

- Membranes are structurally supported by intracellular protein fibers of the cytoskeleton
- Maintains the shape of the membrane



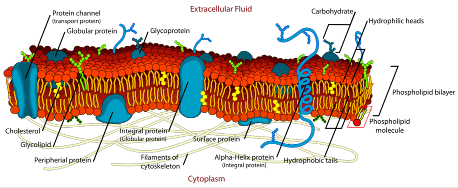
Cell Biology

Biological Membranes

Fluid Mosaic model of the plasma membrane

Exterior proteins and glycolipids

- Proteins and glycerophospholipids on the outside of membrane bears carbohydrate chains – **glycocalyx**
- Glycocalyx acts as cell identity markers



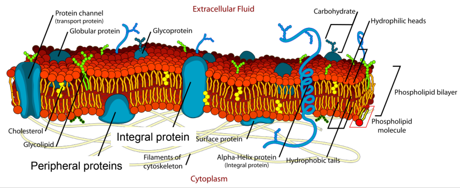
Cell Biology

Biological Membranes

Fluid Mosaic model of the plasma membrane

Membrane proteins

- Float on or in the lipid bilayer
- Most are not fixed in position
- Two types: integral and peripheral membrane proteins
- Provide passageways (transmembrane integral proteins) that allow substances to cross the membrane



Cell Biology

Biological Membranes

Fluid Mosaic model of the plasma membrane

Membrane proteins

- Six different types of membrane proteins
  - Transport proteins/channels
  - Enzymes
  - Cell surface receptors
  - Cell surface identity markers
  - Cell adhesion proteins
  - Cytoskeletal adhesion proteins

Cell Biology

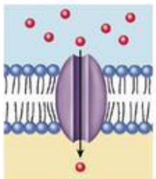
Biological Membranes

Fluid Mosaic model of the plasma membrane

Membrane proteins

Transport proteins/channels

- Large protein molecules that span both phospholipid layers
- Has a hydrophilic interior channel that allows the passage of water soluble molecules (polar and charged molecules)
  e.g. glucose, amino acids, H^+, Na^+, K^+



Cell Biology

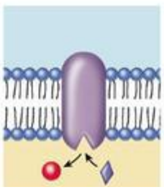
Biological Membranes

Fluid Mosaic model of the plasma membrane

Membrane proteins

Enzymes

- Enzymes are attached to the membrane
- Carry out chemical reactions on the interior and exterior surface of the plasma membrane



Cell Biology

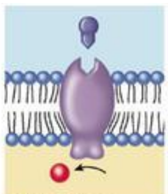
Biological Membranes

Fluid Mosaic model of the plasma membrane

Membrane proteins

Cell surface receptors

- Sensitive to chemical messages from outside
- Also known as receptor proteins – has specific sites to receive specific signal molecules

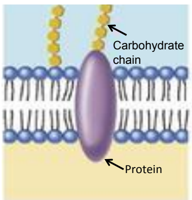


Cell Biology

Biological Membranes
Fluid Mosaic model of the plasma membrane
Membrane proteins

Cell surface identity markers

- Membrane proteins that help identify themselves (i.e. cell) to other cells
- Every cell type has a unique identity
- Made by a specific combination of cell surface proteins
- Usually associated with carbohydrate chains (Glycocalyx)

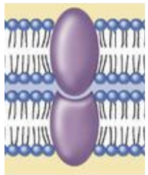


Cell Biology

Biological Membranes
Fluid Mosaic model of the plasma membrane
Membrane proteins

Cell Adhesion proteins

- Proteins that glue cells to each other
- Some make permanent attachments
- Others are temporary

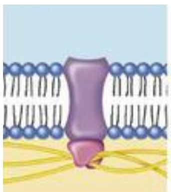


Cell Biology

Biological Membranes
Fluid Mosaic model of the plasma membrane
Membrane proteins

Cytoskeletal adhesion proteins

- Proteins that anchor membrane proteins to the cytoskeleton (actin filaments)



Cell Biology

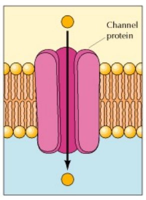
Biological Membranes
Transport of molecules through the membranes

- Happens in two different processes
  - passive: energy not required
  - active: energy required
- Passive transport of molecules through the membranes happens in two methods
  - Diffusion
  - Osmosis

Cell Biology

Biological Membranes
Transport of molecules through the membranes
Passive transport of molecules through the membranes
Diffusion

- Net movement of substances to regions **of lower concentration** as a result of random spontaneous motion
- Membrane transport proteins allow only certain molecules and ions to diffuse through the plasma membrane – **selective permeability**

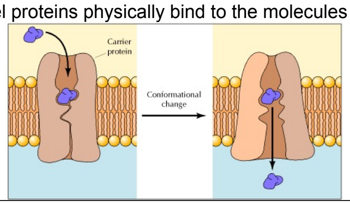


- Cells possess many different kinds of membrane transport proteins for different molecules

Cell Biology

Biological Membranes
Transport of molecules through the membranes
Passive transport of molecules through the membranes
Diffusion

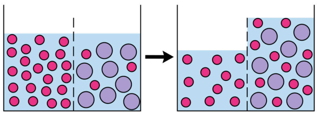
- **Facilitated diffusion** – transport of molecules and ions across a membrane by **specific carriers** in the direction of lower concentration
- Channel proteins physically bind to the molecules



Cell Biology

Biological Membranes
Transport of molecules through the membranes

Passive transport of molecules through the membranes
Osmosis
- Osmosis - **net movement of solvent molecules** (i.e. water in the cell) through a **semi-permeable membrane** (cell membrane) into a region of **higher solute concentration**, in the direction that tends to **equalize the solute concentrations** on the two sides



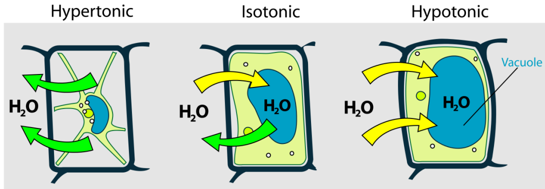
Cell Biology

Biological Membranes
Transport of molecules through the membranes

Passive transport of molecules through the membranes

Osmosis

Hypertonic Isotonic Hypotonic



Cell Biology

Biological Membranes
Transport of molecules through the membranes

Active transport of molecules through the membranes

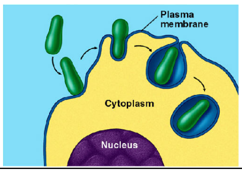
- Method of transport that utilizes energy (i.e. ATP)
- Enables to move solutes across a membrane **against its concentration gradient**
- Three main modes
  - Bulk transport
  - Primary active transport
  - Secondary active transport

Cell Biology

Biological Membranes
Transport of molecules through the membranes

Active transport of molecules through the membranes - Bulk transport
- Transports large amounts of substances across the membrane
  e.g. Endocytosis, Exocytosis

Endocytosis
- Importation of large materials in to cells by engulfing them with their plasma membranes
- If the engulfed material is an organism/organic matter – **phagocytosis**

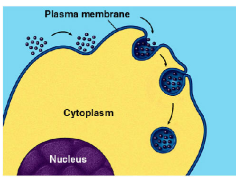


Cell Biology

Biological Membranes
Transport of molecules through the membranes

Active transport of molecules through the membranes - Bulk transport
- Transports large amounts of substances across the membrane
  e.g. Endocytosis, Exocytosis

Endocytosis
- If the engulfed material is in liquid form – **pinocytosis**



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Biological Membranes
Transport of molecules through the membranes

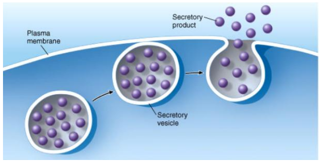
Active transport of molecules through the membranes - Bulk transport
Exocytosis
- Discharge of **large amount of materials out of cells** in the form of **vesicles** at the cell surface
- Important for exporting materials for cell wall formation in plant cells
- In animal cells, exocytosis is a mode for **secretion of hormones, neurotransmitters, digestive enzymes** etc.

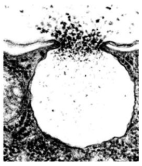
Cell Biology

Biological Membranes
Transport of molecules through the membranes

Active transport of molecules through the membranes - Bulk transport

Exocytosis





Cell Biology

Biological Membranes
Transport of molecules through the membranes

Active transport of molecules through the membranes - Bulk transport

- Energy (as ATP) is utilized during
  - Cytoskeletal rearrangements (vesicle formation and movement)
  - Membrane remodeling (vesicle fusion and budding)
  - Protein function (e.g. SNARE & Calithrin)

Cell Biology

Biological Membranes
Transport of molecules through the membranes

Active transport of molecules through the membranes

Primary active transport

- Directly uses cell's metabolic energy to transport molecules across the membrane (e.g. Sodium-Potassium pump)

Sodium-Potassium pump

- Most animal cells have low internal concentration of Na^+ and high internal concentration of K^+
- Concentration difference is maintained by actively pumping Na^+ out of the cell and pumping in K^+
- Works through a series of conformational changes in the transmembrane proteins

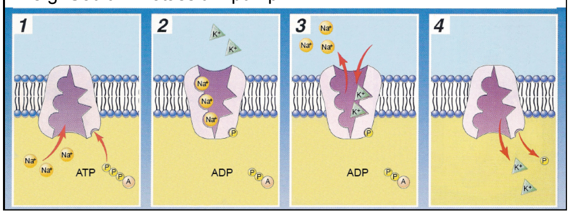
Cell Biology

Biological Membranes
Transport of molecules through the membranes

Active transport of molecules through the membranes

Primary active transport

e.g. Sodium-Potassium pump



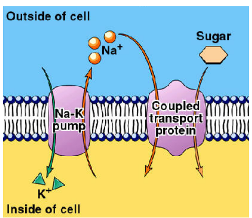
Cell Biology

Biological Membranes
Transport of molecules through the membranes

Active transport of molecules through the membranes

Secondary active transport

- Co-transport molecules against their concentration gradient by coupling their movements with other active transport channels (e.g. Sodium-Potassium pump, proton pump)



Cell Biology

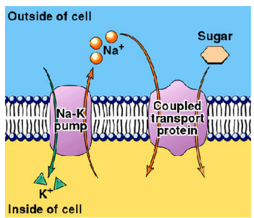
Biological Membranes
Transport of molecules through the membranes

Active transport of molecules through the membranes

Secondary active transport

- Do not use direct energy to transport molecules across the membrane
- Uses electrochemical potential difference across the cell membranes

e.g. Coupled transport channels



Cell Biology

Summary

- Plasma membrane has a fluid mosaic model: mosaic of proteins floating in the fluid lipid bilayer
- Cell membrane comprises of a glycerophospholipid bilayer, membrane proteins, network of supporting fibers and exterior proteins and glycolipids
- Membrane proteins include transport proteins, enzymes, cell surface receptors, cell surface identity markers, cell adhesion proteins, cytoskeletal adhesion proteins
- Transport of molecules through membranes happen through passive and active methods
- Passive transport of molecules through the membranes happens through diffusion and osmosis
- Active transport of molecules through the membranes happens through bulk transport, primary active transport an secondary active transport